Algebra 1 Name____________________
Unit 1 Date____________________
Lesson 4 Practice Period__________________

For questions 1-5, use the exponent laws to write equivalent expressions.

1. $\left(\frac{6}{5}\right)^{-2} =$

2. $\frac{7^7}{7^4} =$

3. $(5.2 \cdot (-9))^3 =$

4. $(-2) \cdot (-2)^{-5} \cdot (-2)^8 =$

5. $((-4)^4)^9 =$

Find the value of each for questions 6-7.

6. $\sqrt[3]{125} =$

7. $\sqrt{25} =$

8. Explain why both the $\sqrt{25}$ and $\sqrt[3]{125}$ both equal 5.

For questions 9-10, find the value of the variable that makes the equation true.

9. $\sqrt{c} \cdot \sqrt{c} = 11$

$c =$ ______

10. $\frac{2^3}{3^3} = \left(\frac{3}{2}\right)^g$

$g =$ ______